

# Steepest Descent Depends on the Metric

## Steepest Among Which Directions?

At a point  $p$ , the differential

$$df_p : T_p M \rightarrow \mathbb{R}$$

tells us the rate of change in every direction:

$$X_p \mapsto df_p(X_p).$$

If we want the direction of fastest decrease, we might try to minimize

$$df_p(X_p)$$

over all nonzero directions  $X_p$ .

But this has no answer unless we say how large a direction is. If  $X_p$  gives some decrease, then  $100X_p$  gives one hundred times as much first-order decrease:

$$df_p(100X_p) = 100 df_p(X_p).$$

So without a notion of unit length, “steepest” is meaningless.

## The Metric Defines Unit Directions

A Riemannian metric gives a norm

$$\|X_p\|_g = \sqrt{g_p(X_p, X_p)}.$$

Now we can ask a meaningful question:

among all  $X_p$  with  $\|X_p\|_g = 1$ , which minimizes  $df_p(X_p)$ ?

Using

$$df_p(X_p) = g_p(\text{grad}_g f(p), X_p),$$

the Cauchy–Schwarz inequality gives

$$df_p(X_p) \geq -\|\text{grad}_g f(p)\|_g \|X_p\|_g.$$

For unit vectors,

$$df_p(X_p) \geq -\|\text{grad}_g f(p)\|_g.$$

Equality occurs when

$$X_p = -\frac{\text{grad}_g f(p)}{\|\text{grad}_g f(p)\|_g}.$$

Thus the negative gradient is the steepest descent direction among unit directions measured by  $g$ .

## No Metric, No Canonical Steepest Direction

On a bare smooth manifold, there is no canonical norm on  $T_p M$ . Therefore there is no canonical notion of unit direction, angle, or steepest descent.

You can still compute directional derivatives:

$$df_p(X_p).$$

But you cannot canonically say which direction is the steepest without choosing some way to measure the size of directions.

## Why Different Metrics Give Different Flows

Changing the metric changes the unit sphere in each tangent space. Since steepest descent means “best decrease per unit length,” changing the notion of length changes the steepest direction.

This is why Euclidean gradient descent, natural gradient descent, Riemannian gradient descent, and Wasserstein gradient flow can all have the same objective but different dynamics: they use different metrics.